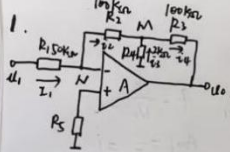


Homework 10.



虚断: $i_N = i_P = 0$. $\bar{v}_1 = \bar{v}_2 = \frac{U_1}{R_1}$ $\bar{v}_3 = -\frac{U_M}{R_3}$

图 $\frac{U_1 - U_M}{R_1} = \frac{U_M - U_M}{R_2} \Rightarrow U_M = -\frac{R_2}{R_1} U_1$

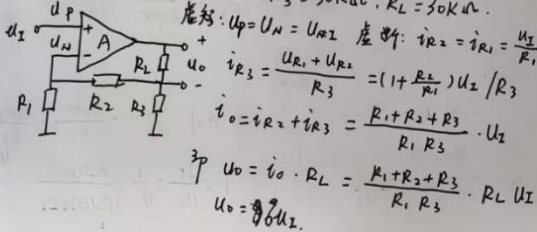
$U_o = U_M - i_4 \cdot R_4$ $i_4 = \bar{v}_2 + \bar{v}_3$

$U_o = -\frac{R_2 + R_4}{R_1} (1 + \frac{R_2}{R_3}) \cdot U_1$

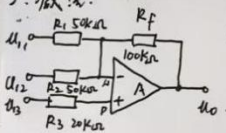
$U_o = -\frac{R_2 R_3}{R_1} (\frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}) \cdot U_1$

代入数据得 $U_o = -10.4 U_1$

2. $R_1 = 10K\Omega$, $R_2 = 20K\Omega$, $R_3 = 30K\Omega$, $R_4 = 30K\Omega$.



3. 设计:



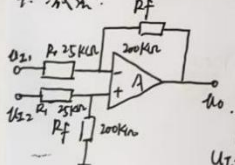
$U_N = U_P = U_{R3}$

节点电压法: $\frac{U_{R1} - U_{R3}}{R_1} + \frac{U_{R2} - U_{R3}}{R_2} = \frac{U_{R3} - U_o}{R_f}$

代入电压值得: $U_o = 5 U_{R3} - 2 U_{R1} - 2 U_{R2}$

$U_o = -\frac{R_f}{R_1} U_{R1} - \frac{R_f}{R_2} U_{R2} + \frac{R_f}{R_3} U_{R3}$

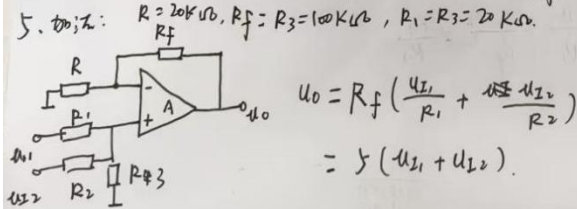
4. 设计:



$U_P = \frac{8}{9} U_{R2}$ $\frac{U_{R1} - U_N}{R_1} = \frac{U_N - U_o}{R_f}$

$U_o = \frac{R_f}{R_1} (U_{R2} - U_{R1}) = 8 (U_{R2} - U_{R1})$

5. 设计:

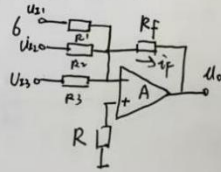


$R = 20K\Omega$, $R_f = R_3 = 100K\Omega$, $R_1 = R_2 = 20K\Omega$.

$U_o = R_f (\frac{U_{R1}}{R_1} + \frac{U_{R2}}{R_2})$

$= 5 (U_{R1} + U_{R2})$

Homework 10.



$R = 20K\Omega, R_f = 200K\Omega, R_1 = R_2 = R_3 = 50K\Omega.$

$$u_0 = -i_f \cdot R_f = -R_f \left(\frac{u_{i1}}{R_1} + \frac{u_{i2}}{R_2} + \frac{u_{i3}}{R_3} \right) \\ = -4(u_{i1} + u_{i2} + u_{i3})$$

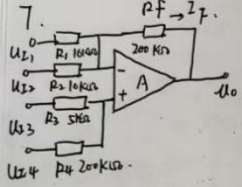
8. ①.

$$u_N = u_P = 0$$

$$i_f = i_{R1} + i_{R2} + i_{R3} \\ = \frac{u_{i1}}{R_1} + \frac{u_{i2}}{R_2} + \frac{u_{i3}}{R_3}$$

$$i_f = \frac{u_N - u_0}{R_f}$$

$$u_0 = -i_f \cdot R_f \\ = -R_f \left(\frac{u_{i1}}{R_1} + \frac{u_{i2}}{R_2} + \frac{u_{i3}}{R_3} \right) \\ = -4(u_{i1} + u_{i2} + u_{i3})$$



$$I_N = I_P = 0, u_N = u_P$$

$$I_f = \frac{u_N - u_0}{R_f} \Rightarrow u_0 = -I_f R_f + u_N$$

$$I_3 + I_4 = I_P = 0 \Rightarrow \frac{u_{i3} - u_P}{R_3} + \frac{u_{i4} - u_P}{R_4} = 0$$

$$I_1 + I_2 = I_N + I_f$$

$$\frac{u_{i1} - u_N}{R_1} + \frac{u_{i2} - u_N}{R_2} = I_f$$

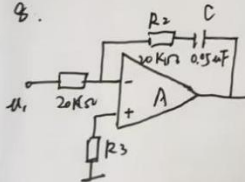
$$\Rightarrow u_0 = - \left(\frac{u_{i1} - u_N}{R_1} + \frac{u_{i2} - u_N}{R_2} \right) R_f + u_N$$

$$u_N = u_P = (R_3 // R_4) \left(\frac{u_{i3}}{R_3} + \frac{u_{i4}}{R_4} \right)$$

$$\Rightarrow u_0 = -R_f \left(\frac{u_{i1}}{R_1} + \frac{u_{i2}}{R_2} \right) + \left(1 + \frac{R_f}{R_1} + \frac{R_f}{R_2} \right) (R_3 // R_4) \left(\frac{u_{i3}}{R_3} + \frac{u_{i4}}{R_4} \right)$$

$$\Rightarrow u_0 = -20u_{i1} - 20u_{i2} + 40u_{i3} + u_{i4}$$

8.



微分积分

$$u_N = u_P = 0$$

$$KCL: \frac{u_i}{R_1} = \frac{-u_0}{R_2 + \frac{1}{sC}}$$

$$\Rightarrow u_0 = -\frac{R_2}{R_1} u_i - \frac{1}{R_1 C} \int u_i dt = -u_i - 1000 \int u_i dt$$